Circuits, Systems, and Signal Processing (2025) 44:7070–7091
https://doi.org/10.1007/s00034-025-03168-9

Parametric Monogenic Linear Canonical Wavelet Transform

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Received: 10 December 2024 / Revised: 27 April 2025 / Accepted: 29 April 2025 /
Published online: 12 May 2025

Abstract

This paper begins by examining the definition and fundamental properties of the two-dimensional linear canonical wavelet transform (2-D LCWT) within the framework of linear canonical transform theory. The LCWT offers significant advantages in handling multi-scale and multi-directional signals. Building on this, a novel model is proposed based on the parametric Riesz transform and parametric monogenic signal, introducing a parametric monogenic linear canonical wavelet along with its associated transform, referred to as PMLW. Through parametric embedding within the monogenic linear canonical wavelet framework, the proposed transform achieves enhanced flexibility and robustness, facilitating more efficient analysis of intricate features in complex signals. This approach leverages 2-D analytical signal theory to incorporate phase information with directional characteristics, thereby preserving rich directional details in multi-scale analysis. Furthermore, the potential applications of PMLW in image denoising tasks are explored, demonstrating its effectiveness in preserving structural details while suppressing noise.

Keywords Linear canonical transform · Analytic wavelet transform · Riesz transform · Monogenic signal · Analytic signal

1 Introduction

The two-dimensional wavelet transform (2-D WT), as a fundamental tool in signal processing, effectively captures the features of signals across multiple scales and directions [3, 10]. It is widely applied in image processing, edge detection, and feature extraction [1, 2, 21]. However, the traditional wavelet transform exhibits certain lim-

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itations when handling complex signals. Its lack of directional sensitivity reduces its effectiveness in characterizing local structures, particularly in applications that require multi-directional analysis or high-frequency texture representation [18, 25]. Moreover, the 2-D WT faces challenges in terms of robustness and translation invariance when exposed to noise, which constrains its performance in practical scenarios [7, 17, 20, 22].

The monogenic wavelet transform (MWT) was developed to address the directional limitations of the traditional 2-D WT, offering an enhanced approach to 2-D signal processing by incorporating phase information alongside the classic multi-scale analysis of the WT [12, 25, 31]. By adding this phase component [11, 15, 29], the MWT enables a more nuanced description of a signal's directional structure, which is particularly advantageous in fields requiring high precision in orientation detection and directional filtering [15, 24, 30, 33]. Unlike the traditional WT, which is generally limited in capturing intricate directional details, the MWT provides richer contextual information within a multi-scale framework, making it a valuable tool for analyzing complex patterns. Moreover, there have been many derivative transformations and practical applications based on MWT, such as significant achievements in image enhancement, texture analysis, and motion estimation [5, 6, 16, 23, 26]. However, while the MWT enhances the directionality limitations of the traditional wavelet transform, it still encounters challenges in processing non-stationary signals. Furthermore, there remains potential for improvement in the representation and decoupling of complex signals [4, 28].

The linear canonical transform (LCT) is a generalized integral transform that extends and unifies traditional time-frequency analysis methods [32]. It encompasses the Fourier transform, fractional Fourier transform, and Fresnel transform as special cases, making it a powerful tool for signal processing and optical system analysis. Building on the theory of the LCT [19, 27], this article first explores the definition and key properties of the 2-D linear canonical wavelet transform (LCWT). By introducing linear canonical wavelets, this transformation enables more efficient analysis of signals in both the frequency and spatial domains [4, 14]. Compared to the traditional 2-D WT, the linear canonical wavelet offers significant advantages in handling multi-scale and multi-directional signals. Furthermore, this approach provides superior energy concentration and spectral control, demonstrating enhanced adaptability in processing non-stationary signals and complex image structures [28].

To address the anisotropy issue in traditional 2-D Hilbert-based transforms, in [8, 9], we developed the parametric Riesz transform (PRT) by leveraging the irrotational and solenoidal characteristics of three-dimensional harmonic fields and the properties of the LCT. Building upon this foundation, this article further proposes a model based on the PRT and the parametric monogenic signal, developing a parametric monogenic linear canonical wavelet (PMLW) and its corresponding transform using the LCT framework. The core of this approach lies in leveraging 2-D analytical signal theory to enhance the flexibility and stability of transformations through parametric design [11]. This novel transformation retains the phase information and directional advantages of monogenic wavelets [25], offering a more effective tool for processing complex signals. It holds significant potential for practical applications in the future.

In this paper, the 2-D LCWT and the PMLW are proposed and their development is outlined through the following steps. Section 2 provides an overview of the definitions of the 2-D WT, LCT, and parametric monogenic signal. Section 3 introduces the definition of the 2-D LCWT and analyzes its properties. Section 4 presents the parametric monogenic linear canonical wavelet and explores its characteristics. Section 5 proposes the PMLW and examines its properties. Section 6 explores the potential applications of PMLW in image denoising tasks. Finally, Section 7 concludes the paper, summarizing the key findings and contributions.

2 Preliminaries

2.1 Two-dimensional Wavelet Transform

Wavelet transform is a powerful tool for multi-scale signal analysis, widely used in image processing and feature extraction. In the 2-D case, the continuous wavelet transform (CWT) of a function $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$ is defined by selecting an admissible mother wavelet $\psi(\mathbf{t}) \in L^2(\mathbb{R}^2)$, which must satisfy the following conditions [3]

$$0 < c_\psi = (2\pi)^2 \int_{\mathbb{R}^2} \frac{|\hat{\psi}(\boldsymbol{\omega})|}{|\boldsymbol{\omega}|^2} d\boldsymbol{\omega} < \infty, \tag{1}$$

$$\int_{\mathbb{R}^2} |\psi(\mathbf{t})|^2 d\mathbf{t} = 1, \tag{2}$$

where $\hat{\psi}(\boldsymbol{\omega})$ represents the Fourier transform of $\psi(\mathbf{t})$, and $|\boldsymbol{\omega}|^2 = \omega_1^2 + \omega_2^2$ corresponds to the squared magnitude of the frequency variable $\boldsymbol{\omega} = (\omega_1, \omega_2)$. Consider the following operators:

- Translation operator: $\mathcal{T}_{\mathbf{l}}\psi(\mathbf{t}) = \psi(\mathbf{t} - \mathbf{l})$.
- Dilation operator: $\mathcal{D}_s\psi(\mathbf{t}) = s^{-1}\psi(\mathbf{t}/s)$.
- Rotation operator: $\mathcal{R}_\theta\psi(\mathbf{t}) = \psi(\mathbf{R}_{-\theta}\mathbf{t})$, where $\mathbf{R}_\theta = ((\cos\theta \; -\sin\theta), (\sin\theta \; -\cos\theta))$, and $\theta \in [0, 2\pi)$.

From the mother wavelet $\psi(\mathbf{t})$, a family of functions $\{\psi_{\boldsymbol{\xi}}(\mathbf{t})\}$ is generated through the following transform:

$$\psi_{\boldsymbol{\xi}}(\mathbf{t}) = \mathcal{D}_s\mathcal{R}_\theta\mathcal{T}_{\mathbf{l}}\psi(\mathbf{t}) = \mathcal{U}_{\boldsymbol{\xi}}\psi(\mathbf{t}), \tag{3}$$

where $\boldsymbol{\xi} = (s, \theta, \mathbf{l})$ represents the set of parameters for dilation (s), rotation (θ), and translation ($\mathbf{l}$). Including the rotation parameter θ enables the wavelet to capture local features, such as unidirectional or separable structures, that are not aligned with the observational axes. By rotating the analyzing function, it ensures flexibility in detecting features at various orientations [3].

The wavelet coefficients of a function $f(\mathbf{t})$ are defined as follows [3]:

$$w_\psi(\boldsymbol{\xi}, f) = \langle \psi_{\boldsymbol{\xi}}(\mathbf{t}), f(\mathbf{t}) \rangle = \int_{\mathbb{R}^2} f(\mathbf{t})\psi_{\boldsymbol{\xi}}^*(\mathbf{t})\, d^2\mathbf{t}, \tag{4}$$

where $*$ denotes the complex conjugate. These wavelet coefficients, $w_\psi(\boldsymbol{\xi}, f)$, enable the reconstruction of the original function $f(\mathbf{t})$ by taking a weighted sum of the wavelet family $\{\psi_{\boldsymbol{\xi}}(\mathbf{t})\}$, expressed as:

$$f(\mathbf{t}) = \frac{1}{c_\psi} \int_{\mathbb{R}^2} \int_0^{2\pi} \int_0^{\infty} w_\psi(\boldsymbol{\xi}, f)\psi_{\boldsymbol{\xi}}(\mathbf{t}) \frac{1}{s^3}\, ds\, d\theta\, d^2\mathbf{l}. \tag{5}$$

2.2 Linear Canonical Transform

The linear canonical transform (LCT), proposed by Moshinsky and Collins in the 1970s, is a generalized integral transform that unifies and extends traditional time-frequency analysis methods [13, 35]. It encompasses the Fourier transform, fractional Fourier transform, and Fresnel transform as special cases, thereby providing a broader framework for signal processing. By extending the time-frequency domain to the LCT domain, it plays a significant role in various applications. Among its various formulations, the 2-D LCT is particularly useful for processing multidimensional signals and images. It is mathematically defined as follows [27]

$$L_{\mathbf{A}}\{f\}(\omega_1, \omega_2) = \int_{\mathbb{R}^2} f(t_1, t_2)\, K_{A_1}(t_1, \omega_1)\, K_{A_2}(t_2, \omega_2)\, \mathrm{d}\mathbf{t}, \tag{6}$$

where$A_k = (a_k, b_k; c_k, d_k)$ satisfies $a_k d_k - b_k c_k = 1$ with $b_k \neq 0$ for $k = 1, 2$, and the kernel $K_{A_k}(t_k, \omega_k)$ is given by

$$K_{A_k}(t_k, \omega_k) = \frac{1}{\sqrt{\mathbf{i}2\pi b_k}}\, e^{\mathbf{i}\left(\frac{a_k}{2b_k}t_k^2 - \frac{1}{b_k}t_k\omega_k + \frac{d_k}{2b_k}\omega_k^2\right)}. \tag{7}$$

Building upon the concept of the 2-D LCT, its extension to the quaternion domain leads to the quaternion linear canonical transform (QLCT). This extension allows for the processing of quaternion-valued signals, making it particularly useful in applications such as color image processing and vector-field analysis. The QLCT of a function $f \in L^1(\mathbb{R}^2; \mathbb{H})$ is similarly defined as [19, 32, 35]

$$L_{q,\mathbf{A}}\{f\}(\omega_1, \omega_2) = \int_{\mathbb{R}^2} K_{\mathbf{i},A_1}(t_1, \omega_1)\, f(t_1, t_2)\, K_{\mathbf{j},A_2}(t_2, \omega_2)\, \mathrm{d}\mathbf{t}, \tag{8}$$

where $\mathbf{I} = (\mathbf{i}, \mathbf{j})$, $\mathbf{A} = (A_1, A_2)$, and the kernel $K_{\mathbf{I}_k,A_k}(t_k, \omega_k)$ is given by

$$K_{\mathbf{I}_k,A_k}(t_k, \omega_k) = \frac{1}{\sqrt{\mathbf{I}_k 2\pi b_k}}\, e^{\mathbf{I}_k\left(\frac{a_k}{2b_k}t_k^2 - \frac{1}{b_k}t_k\omega_k + \frac{d_k}{2b_k}\omega_k^2\right)}. \tag{9}$$

Here, $\mathbf{I}_k$ $(k = 1, 2)$ corresponds to the k-th quaternion imaginary unit selected from $\mathbf{I} = (\mathbf{i}, \mathbf{j})$. Specifically, for the first dimension $(k = 1)$, we set $\mathbf{I}_1 = \mathbf{i}$, and for the second dimension $(k = 2)$, we set $\mathbf{I}_2 = \mathbf{j}$.

2.3 Parametric Monogenic Signal

Existing research has extended the parametric Hilbert transform and total Hilbert transform to the linear canonical domain [19], but these methods remain anisotropic, limiting their effectiveness in multi-dimensional signal analysis. To overcome this limitation, in [9], we developed a parametric Riesz transform (PRT) by leveraging the irrotational and solenoidal characteristics of three-dimensional harmonic fields and the properties of the LCT. This work also introduced the parametric monogenic signal (PMS), providing a more isotropic approach to signal processing in the linear canonical domain.

We define the parametric Riesz kernels as

$$r_{\mathbf{A}}^{(1)}(\boldsymbol{\omega}) = \mathbf{i}\frac{\omega_1/b_1}{|\boldsymbol{\omega}/\mathbf{b}|}, \quad r_{\mathbf{A}}^{(2)}(\boldsymbol{\omega}) = \mathbf{j}\frac{\omega_2/b_2}{|\boldsymbol{\omega}/\mathbf{b}|}, \tag{10}$$

where $\boldsymbol{\omega}/\mathbf{b} = (\omega_1/b_1, \omega_2/b_2)^\top$. The PRT is defined as [9]

$$R_{\mathbf{A}}^{(1)} f(\mathbf{t}) = f_{R_{\mathbf{A}}^{(1)}}(\mathbf{t}) = e^{-\mathbf{i}\frac{a_1}{2b_1}t_1^2} e^{-\mathbf{i}\frac{a_2}{2b_2}t_2^2} \left[f(\mathbf{t}) e^{\mathbf{i}\frac{a_1}{2b_1}t_1^2} e^{\mathbf{i}\frac{a_2}{2b_2}t_2^2} * \frac{t_1}{2\pi |\mathbf{t}|^3} \right], \tag{11}$$

$$R_{\mathbf{A}}^{(2)} f(\mathbf{t}) = f_{R_{\mathbf{A}}^{(2)}}(\mathbf{t}) = e^{-\mathbf{i}\frac{a_1}{2b_1}t_1^2} e^{-\mathbf{i}\frac{a_2}{2b_2}t_2^2} \left[f(\mathbf{t}) e^{\mathbf{i}\frac{a_1}{2b_1}t_1^2} e^{\mathbf{i}\frac{a_2}{2b_2}t_2^2} * \frac{t_2}{2\pi |\mathbf{t}|^3} \right], \tag{12}$$

where $*$ denotes the 2-D convolution. The PMS and parametric anti-monogenic signal (PAMS) of real signal $f(\mathbf{t})$ are defined by

$$\begin{aligned} f_{\mathbf{A}}^{\pm}(\mathbf{t}) = \mathcal{M}_{\mathbf{A}}^{\pm} f(\mathbf{t}) &= f(\mathbf{t}) \mp \mathbf{i} R_{\mathbf{A}}^{(1)} f(\mathbf{t}) \mp \mathbf{j} R_{\mathbf{A}}^{(2)} f(\mathbf{t}) && (13) \\ &= f(\mathbf{t}) \mp \mathbf{i} f_{R_{\mathbf{A}}^{(1)}}(\mathbf{t}) \mp \mathbf{j} f_{R_{\mathbf{A}}^{(2)}}(\mathbf{t}). && (14) \end{aligned}$$

This dual formulation enables a more complete characterization of signal features, particularly in cases where symmetric or anti-symmetric structures need to be analyzed. Furthermore, signal analysis often requires invariance to orientation or the ability to analyze directional variations. To achieve this, we introduce the rotated parametric monogenic signal (RPMS) and the rotated parametric anti-monogenic signal (RPAMS). By applying a rotation transformation, these signals provide a more flexible framework for handling directional information [25]. The RPMS and RPAMS are defined as

$$\begin{aligned} f_{\mathbf{A},\theta}^{\pm}(\mathbf{t}) = \mathcal{R}_\theta \mathcal{M}_{\mathbf{A}}^{\pm} f(\mathbf{t}) &= f(\mathbf{R}_{-\theta}\mathbf{t}) \mp \mathbf{i} \mathcal{R}_\theta R_{\mathbf{A}}^{(1)} f(\mathbf{t}) \mp \mathbf{j} \mathcal{R}_\theta R_{\mathbf{A}}^{(2)} f(\mathbf{t}) \\ &= f(\mathbf{R}_{-\theta}\mathbf{t}) \mp \mathbf{i} f_{R_{\mathbf{A}}^{(1)}}(\mathbf{R}_{-\theta}\mathbf{t}) \mp \mathbf{j} f_{R_{\mathbf{A}}^{(2)}}(\mathbf{R}_{-\theta}\mathbf{t}) \\ &= f_\theta(\mathbf{t}) \mp \mathbf{i} f_{R_{\mathbf{A}}^{(1)},\theta}(\mathbf{t}) \mp \mathbf{j} f_{R_{\mathbf{A}}^{(2)},\theta}(\mathbf{t}). \end{aligned} \tag{15}$$

3 Two-dimentional Linear Canonical Wavelet Transform

Y. Guo extended the one-dimensional linear canonical wavelet transform (LCWT) [4] to two dimensions in [14]. However, the rotation factor was not taken into account. To address this limitation, this paper first analyzes the 2-D LCWT with rotation factors, providing a crucial foundation for the development of the subsequent parametric monogenic linear canonical wavelet transform.

The generalized translation in LCT domain is defined as

$$\begin{aligned}
&\psi_{A_s}(\mathbf{t}\Theta\mathbf{l})\\
&= \int_{\mathbb{R}^2} K^*_{A_1}(l_1,\omega_1)\, K_{A_1}(t_1,\omega_1)\, L_{\mathbf{A}}\{\psi(\mathbf{t})\}(\boldsymbol{\omega}) K_{A_2}(t_2,\omega_2)\, K^*_{A_1}(l_2,\omega_2)\, d\boldsymbol{\omega}\\
&= e^{\frac{\mathbf{i}a_1}{2b_1}(t_1^2-l_1^2)}\left[\frac{1}{(2\pi)^2 b_1 b_2}\int_{\mathbb{R}^2} e^{\frac{\mathbf{i}(l_1-t_1)\omega_1}{b_1}} L_{\mathbf{A}}\{\psi(\mathbf{t})\}(\boldsymbol{\omega}) e^{\frac{\mathbf{i}(l_2-t_2)\omega_2}{b_2}}\, d\boldsymbol{\omega}\right]\\
&\quad\times e^{\frac{\mathbf{i}a_2}{2b_2}(t_2^2-l_2^2)}.
\end{aligned} \tag{16}$$

We introduce a new family of analyzing functions based on the fundamental expression (3) and (16). Let $s\in\mathbb{R}^+$, $\mathbf{I}\in\mathbb{R}^2$, and $\mathbf{R}_\theta\in \mathrm{SO}(2)$, $0<\theta<2\pi$, where SO(2) is the special orthogonal group of rotation in $\mathbb{R}^2$. We define a unitary operator $\mathcal{U}_{\boldsymbol{\xi},\mathbf{A}}$, with the parameters $\boldsymbol{\xi}=(s,\theta,\mathbf{l})$. For any function $\psi(\mathbf{t})\in L^2(\mathbb{R}^2)$, the linear canonical wavelets are given by

$$\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) = \mathcal{U}_{\boldsymbol{\xi},\mathbf{A}}\psi(\mathbf{t}) = s^{-1}\psi\left(\mathbf{R}_{-\theta}\frac{\mathbf{t}-\mathbf{l}}{s}\right) e^{-\mathbf{i}\frac{a_1}{2b_1}(t_1^2-l_1^2)} e^{-\mathbf{i}\frac{a_2}{2b_2}(t_2^2-l_2^2)}, \tag{17}$$

where $A_k=(a_k,b_k;c_k,d_k)$, $k=1,2$. Additionally, it is crucial to examine the 2-D LCT of the linear canonical wavelet. To this end, we present the following lemma for further analysis.

Lemma 1 *For any $\psi(\mathbf{t})\in L^2(\mathbb{R}^2)$, we have*

$$\begin{aligned}
L_{\mathbf{A}}\{\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})\}(\boldsymbol{\omega}) &= \frac{s}{2\pi\sqrt{-b_1b_2}} e^{\mathbf{i}\frac{a_1l_1^2-2l_1\omega_1+d_1\omega_1^2-d_1s^2\omega_1^2}{2b_1}} e^{\mathbf{i}\frac{a_2l_2^2-2l_2\omega_2+d_1\omega_2^2-d_2s^2\omega_2^2}{2b_2}}\\
&\quad\times L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}).
\end{aligned} \tag{18}$$

Proof From the definition of the 2-D LCT in (6), the following result can be directly obtained:

$$\begin{aligned}
L_{\mathbf{A}}\{\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})\}(\boldsymbol{\omega}) &= \frac{1}{2\pi\sqrt{-b_1b_2}}\int_{\mathbb{R}^2} s^{-1}\psi\left(\mathbf{R}_{-\theta}\frac{\mathbf{t}-\mathbf{l}}{s}\right) e^{-\mathbf{i}\frac{a_1}{2b_1}(t_1^2-l_1^2)} e^{-\mathbf{i}\frac{a_2}{2b_2}(t_2^2-l_2^2)}\\
&\quad\times e^{\mathbf{i}\left(\frac{a_1}{2b_1}t_1^2-\frac{1}{b_1}t_1\omega_1+\frac{d_1}{2b_1}\omega_1^2\right)} e^{\mathbf{i}\left(\frac{a_2}{2b_2}t_2^2-\frac{1}{b_2}t_2\omega_2+\frac{d_2}{2b_2}\omega_2^2\right)} d\mathbf{t}\\
&= \frac{1}{2\pi s\sqrt{-b_1b_2}} e^{\mathbf{i}\frac{1}{2b_1}(a_1l_1^2+d_1\omega_1^2)} e^{\mathbf{i}\frac{1}{2b_2}(a_2l_2^2+d_2\omega_2^2)} \int_{\mathbb{R}^2}\psi\left(\mathbf{R}_{-\theta}\frac{\mathbf{t}-\mathbf{l}}{s}\right)
\end{aligned}$$

$$\times\, e^{-\mathbf{i}\frac{1}{b_1}t_1\omega_1} e^{-\mathbf{i}\frac{1}{b_2}t_2\omega_2} d\mathbf{t}. \tag{19}$$

By applying the variable substitution $\mathbf{x} = \frac{\mathbf{t}-\mathbf{l}}{s}$, the expression is transformed as

$$\begin{aligned}
L_{\mathbf{A}}\{\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})\}(\boldsymbol{\omega}) &= \frac{s}{2\pi\sqrt{-b_1b_2}} e^{\mathbf{i}\frac{1}{2b_1}(-2l_1\omega_1+d_1\omega_1^2)} e^{\mathbf{i}\frac{1}{2b_2}(-2l_2\omega_2+d_1\omega_2^2)} \\
&\quad\times \int_{\mathbb{R}^2} \psi\left(\mathbf{R}_{-\theta}\mathbf{x}\right) e^{\mathbf{i}\frac{1}{2b_1}(a_1l_1^2-2x_1s\omega_1)} e^{\mathbf{i}\frac{1}{2b_2}(a_2l_2^2-2x_2\omega_2)} d\mathbf{x}. \\
&= \frac{s}{2\pi\sqrt{-b_1b_2}} e^{\mathbf{i}\frac{1}{2b_1}(a_1l_1^2-2l_1\omega_1+d_1\omega_1^2-d_1s^2\omega_1^2)} e^{\mathbf{i}\frac{1}{2b_2}(a_2l_2^2-2l_2\omega_2+d_1\omega_2^2-d_2s^2\omega_2^2)} \\
&\quad\times \int_{\mathbb{R}^2} \psi\left(\mathbf{R}_{-\theta}\mathbf{x}\right) e^{-\mathbf{i}\frac{a_1}{2b_1}x_1^2} e^{-\mathbf{i}\frac{a_2}{2b_2}x_2^2} e^{\mathbf{i}\frac{1}{2b_1}(a_1x_1^2-2x_1s\omega_1+d_1s^2\omega_1^2)} e^{\mathbf{i}\frac{1}{2b_2}(a_2x_2^2-2x_2\omega_2+d_2s^2\omega_2^2)} d\mathbf{x}. \\
&= \frac{s}{2\pi\sqrt{-b_1b_2}} e^{\mathbf{i}\frac{1}{2b_1}(a_1l_1^2-2l_1\omega_1+d_1\omega_1^2-d_1s^2\omega_1^2)} e^{\mathbf{i}\frac{1}{2b_2}(a_2l_2^2-2l_2\omega_2+d_1\omega_2^2-d_2s^2\omega_2^2)} \\
&\quad\times L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}).
\end{aligned} \tag{20}$$

□

A key characteristic of the linear canonical wavelets defined in (17) is the following admissibility condition.

Definition 1 A mother wavelet $\psi(\mathbf{t}) \in L^2(\mathbb{R}^2)$, associated with the 2-D LCT, is considered admissible if and only if it satisfies the following condition

$$0 < c_{\psi,\mathbf{A}} = \int_{\mathbb{R}^+\times \mathrm{SO}(2)} \left|L_{\mathbf{A}}\left\{\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})\right\}(\boldsymbol{\omega})\right|^2 \frac{ds d\theta}{s^3} < \infty, \quad \text{a.e. } \boldsymbol{\omega} \in \mathbb{R}^2. \tag{21}$$

Example 1 Consider the difference-of-Gaussian (DOG) wavelets, defined by the function [28]

$$\psi(\mathbf{t}) = \lambda^{-2} e^{\frac{-1}{2\lambda^2}(t_1^2+t_2^2)} - e^{\frac{-1}{2}(t_1^2+t_2^2)}, \quad 0 < \lambda < 1. \tag{22}$$

For the case where $\theta = 0$, implying that the rotation matrix $\mathbf{R}_{-\theta}$ is the identity matrix I, the corresponding family of linear canonical wavelets is expressed as

$$\begin{aligned}
\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) =& s^{-1}\left[\lambda^{-2} e^{\frac{-1}{2\lambda^2 s^2}\left((t_1-l_1)^2+(t_2-l_2)^2\right)} - e^{\frac{-1}{2s^2}\left((t_1-l_1)^2+(t_2-l_2)^2\right)}\right] \\
&\times e^{-\mathbf{i}\frac{a_1}{2b_1}(t_1^2-l_1^2)} e^{-\mathbf{i}\frac{a_2}{2b_2}(t_2^2-l_2^2)}.
\end{aligned} \tag{23}$$

We first examine how scale and translation affect the wavelet structure. As shown in Fig. 1, increasing the scale s broadens the wavelet while reducing its peak amplitude, whereas a smaller s localizes it with higher intensity. The translation parameters l_1 and l_2 shift the wavelet spatially, enabling feature localization at different positions.

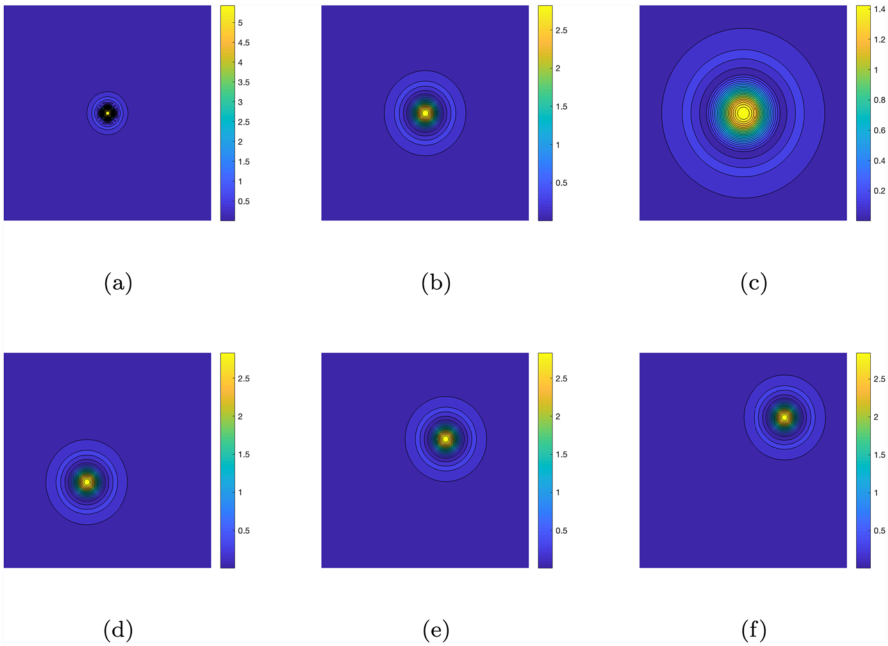

Fig. 1 Comparison of different scales and translations. (a) $s = 0.5, l_1 = l_2 = 0$. (b) $s = 1, l_1 = l_2 = 0$. (c) $s = 2, l_1 = l_2 = 0$. (d) $s = 1, l_1 = l_2 = -1$. (e) $s = 1, l_1 = l_2 = 1$. (f) $s = 1, l_1 = l_2 = 2$

We then analyze the impact of linear canonical parameters, as shown in Fig. 2. The real and imaginary components (Fig. 2(d)-(i)) illustrate phase modulation effects. Increasing a_k and b_k enhances phase oscillations, altering the wavelet?s time-frequency characteristics. This provides additional flexibility for adaptive time-frequency representation.

The formal definition of the 2-D LCWT can now be introduced.

Definition 2 Let $\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) \in L^2(\mathbb{R}^2)$ be a linear canonical wavelet. For any function $f(\mathbf{t}) \in L^2(\mathbb{R}^2)$, the 2-D linear canonical wavelet transform is defined as

$$w_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f) = \int_{\mathbb{R}^2} f(\mathbf{t})\psi^*_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})\, d\mathbf{t}, \tag{24}$$

where $\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})$ is provided by (17).

The LCWT can be rewritten in terms of the LCT, which is described in the following lemma.

Lemma 2 *Suppose that $f(\mathbf{t}), \psi(\mathbf{t}) \in L^2(\mathbb{R}^2)$. The LCT of the LCWT is given by*

$$\begin{aligned} &L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}\right\} \\ &= s e^{\mathbf{i}\frac{d_1 s^2 \omega_1^2}{2b_1}} e^{\mathbf{i}\frac{d_2 s^2 \omega_2^2}{2b_2}} L_{\mathbf{A}}\{f\}(\boldsymbol{\omega}) L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^* (s\mathbf{R}_{-\theta}\boldsymbol{\omega}). \end{aligned} \tag{25}$$

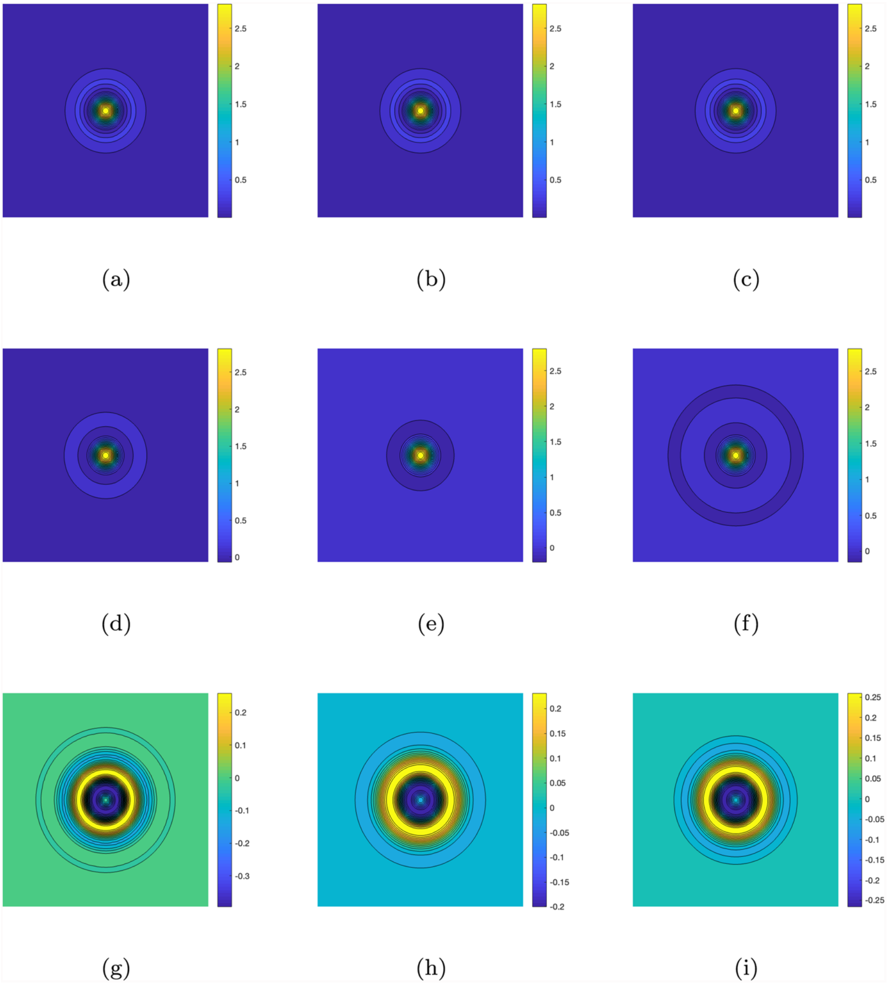

Fig. 2 Comparison of different linear canonical parameters. (a) Modulus with $a_1 = a_2 = 1, b_1 = b_2 = 0.5$. (b) Modulus with $a_1 = a_2 = 1, b_1 = b_2 = 1$. (c) Modulus with $a_1 = a_2 = 2, b_1 = b_2 = 1.5$. (d) Real-part with $a_1 = a_2 = 1, b_1 = b_2 = 0.5$. (e) Real-part with $a_1 = a_2 = 1, b_1 = b_2 = 1$. (f) Real-part with $a_1 = a_2 = 2, b_1 = b_2 = 1.5$. (g) Imaginary-part with $a_1 = a_2 = 1, b_1 = b_2 = 0.5$. (h) Imaginary-part with $a_1 = a_2 = 1, b_1 = b_2 = 1$. (i) Imaginary-part with $a_1 = a_2 = 2, b_1 = b_2 = 1.5$

Proof We have

$$\begin{aligned} w_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f) &= \left(f, \psi_{\boldsymbol{\xi},\mathbf{A}}\right)_{L^2(\mathbb{R}^2)} \\ &= \left(L_{\mathbf{A}}\{f\}, L_{\mathbf{A}}\left\{\psi_{\boldsymbol{\xi},\mathbf{A}}\right\}\right)_{L^2(\mathbb{R}^2)} \\ &= \int_{\mathbb{R}^2} L_{\mathbf{A}}\{f\}(\boldsymbol{\nu}) L_{\mathbf{A}}\left\{\psi_{\boldsymbol{\xi},\mathbf{A}}\right\}^*(\boldsymbol{\omega})\, d\boldsymbol{\omega}. \end{aligned}$$

Applying Lemma 1, we obtain

$$\begin{aligned} w_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f) =& \frac{s}{2\pi\sqrt{-b_1 b_2}} e^{-\mathbf{i}\frac{a_1 l_1^2 - 2l_1\omega_1 + d_1\omega_1^2 - d_1 s^2\omega_1^2}{2b_1}} e^{-\mathbf{i}\frac{a_2 l_2^2 - 2l_2\omega_2 + d_1\omega_2^2 - d_2 s^2\omega_2^2}{2b_2}} \\ &\times \int_{\mathbb{R}^2} L_{\mathbf{A}}\{f\}(\boldsymbol{\omega}) L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^* (s\mathbf{R}_{-\theta}\boldsymbol{\omega})\, d\boldsymbol{\omega}. \end{aligned}$$

Using the inversion formula for the LCT gives

$$\begin{aligned} w_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f) &= \frac{s e^{\mathbf{i}\frac{d_1 s^2\omega_1^2}{2b_1}} e^{\mathbf{i}\frac{d_2 s^2\omega_2^2}{2b_2}}}{2\pi\sqrt{-b_1 b_2}} \int_{\mathbb{R}^2} L_{\mathbf{A}}\{f\}(\boldsymbol{\omega}) L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^* (s\mathbf{R}_{-\theta}\boldsymbol{\omega}) \\ &\quad \times e^{-\mathbf{i}\frac{a_1 l_1^2 - 2l_1\omega_1 + d_1\omega_1^2}{2b_1}} e^{-\mathbf{i}\frac{a_2 l_2^2 - 2l_2\omega_2 + d_1\omega_2^2}{2b_2}} d\boldsymbol{\omega} \\ &= s L_{\mathbf{A}}^{-1}\left\{ e^{\mathbf{i}\frac{d_1 s^2\omega_1^2}{2b_1}} e^{\mathbf{i}\frac{d_2 s^2\omega_2^2}{2b_2}} L_{\mathbf{A}}\{f\}(\boldsymbol{\omega}) L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^* (s\mathbf{R}_{-\theta}\boldsymbol{\omega})\right\}(\mathbf{l}). \end{aligned}$$

Hence, we have

$$L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}\right\} = s e^{\mathbf{i}\frac{d_1 s^2\omega_1^2}{2b_1}} e^{\mathbf{i}\frac{d_2 s^2\omega_2^2}{2b_2}} L_{\mathbf{A}}\{f\}(\boldsymbol{\omega}) L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^* (s\mathbf{R}_{-\theta}\boldsymbol{\omega}).$$

□

Building on the definitions and properties of the 2-D linear canonical wavelet and the LCWT discussed earlier, further analysis will be conducted on the definition and characteristics of the parametric monogenic linear canonical wavelet and parametric monogenic linear canonical wavelet transform (PMLW).

4 Parametric Monogenic Linear Canonical Wavelet

In this section, the concept of the monogenic linear canonical wavelet will be clearly introduced, along with a detailed exploration of its key properties and their significance.

Definition 3 The parametric monogenic extension of any given linear canonical wavelet $\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})$ is given by

$$\psi^+_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) = \mathcal{M}^+_{\mathbf{A}}\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) = \psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) - \mathbf{i}R^{(1)}_{\mathbf{A}}\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) - \mathbf{j}R^{(2)}_{\mathbf{A}}\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}), \tag{26}$$

where $R^{(1)}_{\mathbf{A}}$, $R^{(2)}_{\mathbf{A}}$ are defined in (11) and (12).

To understand the properties of the parametric monogenic linear canonical wavelet, we derive the form of the LCT representations of the wavelets, as shown in Lemma 3.

Lemma 3 *The 2-D LCT of $R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})$, $R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})$ and $\psi_{\xi,\mathbf{A}}^{+}(\mathbf{t})$ is*

$$L_{\mathbf{A}}\{R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\} = \mathbf{i}\cos(\phi-\theta)L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}\}, \tag{27}$$

$$L_{\mathbf{A}}\{R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\} = \mathbf{i}\sin(\phi-\theta)L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}\}, \tag{28}$$

$$L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}^{+}(\mathbf{t})\} = (1+\cos(\phi-\theta)-\mathbf{k}\sin(\phi-\theta)))L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\}, \tag{29}$$

where $\phi = \tan^{-1}\left(\frac{\omega_2/b_2}{\omega_1/b_1}\right)$.

Proof Based on the derivation in [8], we have

$$L_{\mathbf{A}}\{R_{\mathbf{A}}^{(k)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\}(\boldsymbol{\omega}) = \mathbf{i}\frac{\omega_k/b_k}{|\boldsymbol{\omega}/\mathbf{b}|}L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\}(\boldsymbol{\omega}), \tag{30}$$

where $k = 1, 2$. The rotated coordinates can be written as

$$\mathbf{R}_{-\theta}\mathbf{t} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}\begin{bmatrix} \omega_1/b_1 \\ \omega_2/b_2 \end{bmatrix} = \begin{bmatrix} \omega_1\cos\theta/b_1 + \omega_2\sin\theta/b_2 \\ -\omega_1\cos\theta/b_1 + \omega_2\sin\theta/b_2 \end{bmatrix}.$$

Let $\phi = \tan^{-1}\left(\frac{\omega_2/b_2}{\omega_1/b_1}\right)$, then we have

$$\begin{aligned} L_{\mathbf{A}}\{R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\} &= L_{\mathbf{A}}\{R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{R}_{-\theta}\mathbf{t})\} \\ &= \mathbf{i}\frac{\omega_1\cos\theta/b_1 + \omega_2\sin\theta/b_2}{|\boldsymbol{\omega}/\mathbf{b}|}L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\} \\ &= \mathbf{i}\cos(\phi-\theta)L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}\}, \end{aligned} \tag{31}$$

$$L_{\mathbf{A}}\{R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\} = \mathbf{i}\sin(\phi-\theta)L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}\}. \tag{32}$$

Finally, we obtain

$$\begin{aligned} L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}^{+}(\mathbf{t})\} &= L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\} - \mathbf{i}L_{\mathbf{A}}\{R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\} - \mathbf{j}L_{\mathbf{A}}\{R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\} \\ &= (1+\cos(\phi-\theta)-\mathbf{k}\sin(\phi-\theta)))L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\}. \end{aligned} \tag{33}$$

□

The following outlines several key properties that are of particular importance.

Theorem 1 *Each component of the PRT of a linear canonical wavelet, as well as the parametric monogenic extension of the linear canonical wavelet, also retains the wavelet structure.*

Proof First, consider the following inequality

$$\int_{\mathbb{R}^+\times \mathrm{SO}(2)} \left|L_{\mathbf{A}}\left\{R_{\mathbf{A}}^{(k)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right\}(\boldsymbol{\omega})\right|^2 \frac{ds\,d\theta}{s^3} < \int_{\mathbb{R}^+\times \mathrm{SO}(2)} \left|L_{\mathbf{A}}\left\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\right\}(\boldsymbol{\omega})\right|^2 \frac{ds\,d\theta}{s^3}$$
$$< \infty,$$

where $k = 1, 2$. Thus, since these wavelets are derived from linear canonical wavelets, the condition of square integrability is fulfilled. Let $c_{\psi^{(k)},\mathbf{A}}$ denote the value of (21) for the wavelet $R_{\mathbf{A}}^{(k)}\psi_{\xi,\mathbf{A}}(\mathbf{t})$. Using Lemma 3, along with the fact that $\cos^2\phi$ and $\sin^2\phi$ always lie between 0 and 1, it can be inferred that $0 < c_{\psi^{(k)},\mathbf{A}} < c_{\psi,\mathbf{A}} < \infty$, unless $R_{\mathbf{A}}^{(k)}\psi_{\xi,\mathbf{A}}(\mathbf{t}) = 0$. Therefore, $R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})$ and $R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})$ satisfy the admissibility condition in (21), allowing them to be regarded as valid linear canonical wavelets.

Furthermore, it is evident from (26) and (29) that

$$\left|L_{\mathbf{A}}\{\psi^+_{\xi,\mathbf{A}}(\mathbf{t})\}\right|^2 = \left|L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\}\right|^2 + \left|L_{\mathbf{A}}\{R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\}\right|^2 + \left|L_{\mathbf{A}}\{R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\}\right|^2 + 2\sin\phi\left|L_{\mathbf{A}}\{\psi_{\xi,\mathbf{A}}(\mathbf{t})\}\right|^2. \tag{34}$$

Utilizing the Hermitian property of the PRT for real-valued images, it follows that $c_{\psi^{(+)},\mathbf{A}} = c_{\psi,\mathbf{A}} + c_{\psi^{(1)},\mathbf{A}} + c_{\psi^{(2)},\mathbf{A}}$. Hence, since each of $\psi_{\xi,\mathbf{A}}$, $R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}$, and $R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}$ satisfies the admissibility condition, it can be concluded that $\psi^+_{\xi,\mathbf{A}}$ also satisfies the admissibility condition and can be regarded as a parametric monogenic linear canonical wavelet.

Thus, the parametric monogenic extension of the wavelet remains a valid linear canonical wavelet, and its properties may now be further explored. □

Building on the DOG wavelet from Example 1, we now consider the corresponding parameterized monogenic linear canonical wavelet, leading to the formulation of Example 2 below.

Example 2 For the linear canonical wavelet $\psi_{\xi,\mathbf{A}}(\mathbf{t})$ in (23), the parametric monogenic extension is given by $\psi^+_{\xi,\mathbf{A}}(\mathbf{t}) = \psi_{\xi,\mathbf{A}}(\mathbf{t}) - \mathbf{i}R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t}) - \mathbf{j}R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})$, as shown in Fig. 3. The results in the LCT domain, shown in (d) and (e), demonstrate that the parametric monogenic extension effectively suppresses negative frequencies, leading to a more structured and directional representation in the transformed domain.

The parametric monogenic linear canonical wavelet can be expressed in polar form through its modulus, orientation, and phase, respectively, as follows:

$$\psi^+_{\xi,\mathbf{A}}(\mathbf{t}) = \left|\psi^+_{\xi,\mathbf{A}}(\mathbf{t})\right| e^{2\pi\mathbf{e}_{\xi,\mathbf{A}}(\mathbf{t})\phi_{\xi,\mathbf{A}}(\mathbf{t})}, \tag{35}$$

$$\left|\psi^+_{\xi,\mathbf{A}}(\mathbf{t})\right| = \sqrt{\psi^2_{\xi,\mathbf{A}}(\mathbf{t}) + \left(R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2 + \left(R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2}, \tag{36}$$

$$\begin{aligned}\mathbf{e}_{\xi,\mathbf{A}}(\mathbf{t}) &= \frac{\mathbf{i}R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})}{\sqrt{\left(R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2 + \left(R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2}} + \frac{\mathbf{j}R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})}{\sqrt{\left(R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2 + \left(R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2}}\\ &= \mathbf{i}\cos(\upsilon_{\xi,\mathbf{A}}(\mathbf{t})) + \mathbf{j}\sin(\upsilon_{\xi,\mathbf{A}}(\mathbf{t})),\end{aligned} \tag{37}$$

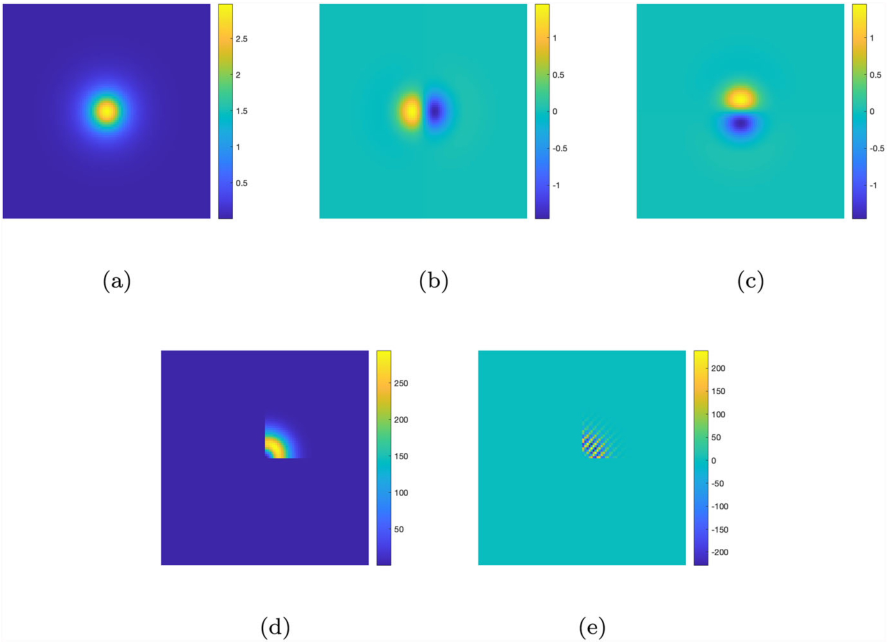

Fig. 3 The parametric monogenic extension of the linear canonical wavelet in (23) with $\lambda = 0.5$, and $A_1 = A_2 = (0, 2, -1/2, 0)$, along with its LCT. (a) Real-part of the parametric monogenic extension. (b) **i**-part of the parametric monogenic extension. (c) **j**-part of the parametric monogenic extension. (d) Real-part of the LCT. (e) **k**-part of the LCT

$$\phi_{\xi,\mathbf{A}}(\mathbf{t}) = \frac{\tau_{\xi,\mathbf{A}}(\mathbf{t})}{2\pi} \tan^{-1}\left(\frac{\sqrt{\left(R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2 + \left(R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\right)^2}}{\psi_{\xi,\mathbf{A}}(\mathbf{t})} \right), \tag{38}$$

where

$$v_{\xi,\mathbf{A}}(\mathbf{t}) = \tan\left(\frac{R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})}{R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})} \right), \tag{39}$$

$$\tau_{\xi,\mathbf{A}}(\mathbf{t}) = \operatorname{sgn}\left((\cos(v_{\xi,\mathbf{A}}(\mathbf{t}))\ \ \sin(v_{\xi,\mathbf{A}}(\mathbf{t}))^{\top} (R_{\mathbf{A}}^{(1)}\psi_{\xi,\mathbf{A}}(\mathbf{t})\ \ R_{\mathbf{A}}^{(2)}\psi_{\xi,\mathbf{A}}(\mathbf{t})) \right). \tag{40}$$

Example 3 Based on the results from Example 2, the corresponding modulus, orientation, and phase are computed, as shown in Fig. 4. The orientation represents the real projection of the quaternion direction vector, providing an intuitive indication of the signal's main direction.

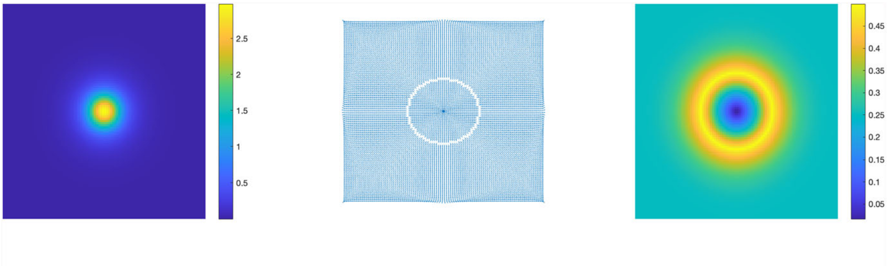

Fig. 4 Polar form of the parametric monogenic linear canonical wavelet. (a) Modulus. (b) Orientation. (c) Phase

5 Parametric Monogenic Linear Canonical Wavelet Transform

If a non-stationary signal $f(\mathbf{t})$ consists of a single component, its local features can be effectively extracted using the parametric monogenic extension. In such cases, the polar representation of Equation (35) can be directly applied to describe the signal. However, most signals exhibit a multiscale structure at a given spatial point, necessitating both positional and scale localization for accurate characterization. To achieve meaningful polar representations under these conditions, it is crucial to employ tools such as the parametric monogenic linear canonical wavelet, which provides the required precision for capturing information in both the spatial and frequency domains. The following provides the definition of the parametric monogenic linear canonical wavelet transform (PMLW).

Definition 4 For the parametric monogenic linear canonical wavelet $\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})$, the PMLW coefficients of a signal $f(\mathbf{t})$ are given by

$$\begin{aligned} w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f) &= \int_{\mathbb{R}^2} f(\mathbf{t})(\psi^{+}_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}))^{*}\, d^2\mathbf{t} \\ &= w_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f) + \mathbf{i} w^{(1)}_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f) + \mathbf{j} w^{(2)}_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f). \end{aligned} \tag{41}$$

The computational complexity of the PMLW computation consists of three main steps. The first step, which involves generating the linear canonical wavelet through modulation and scaling, has a complexity of $O(N^2)$. Next, the parametric Riesz transforms require filtering for two components, contributing a complexity of $2 \times O(N^2 \log N)$. Finally, the inner product integration is computed using convolution, resulting in a complexity of $O(N^2 \log N)$. Considering the dominant terms, the overall complexity of PMLW computation is $O(N^2 \log N)$.

Theorem 2 *The LCT of the PMLW of the signal is expressed as*

$$\begin{aligned} L_{\mathbf{A}}\{w^{+}_{\psi,\mathbf{A}}\}(\boldsymbol{\zeta}) &= L_{\mathbf{A}}\{w_{\psi,\mathbf{A}}\}(\boldsymbol{\zeta}) + \mathbf{i} L_{\mathbf{A}}\{w^{(1)}_{\psi,\mathbf{A}}\}(\boldsymbol{\zeta}) + \mathbf{j} L_{\mathbf{A}}\{w^{(2)}_{\psi,\mathbf{A}}\}(\boldsymbol{\zeta}) \\ &= (1 + \cos(\phi_l - \theta) - \mathbf{k}\sin(\phi_l - \theta))\, L_{\mathbf{A}}\{w_{\psi,\mathbf{A}}\}(\boldsymbol{\zeta}), \end{aligned} \tag{42}$$

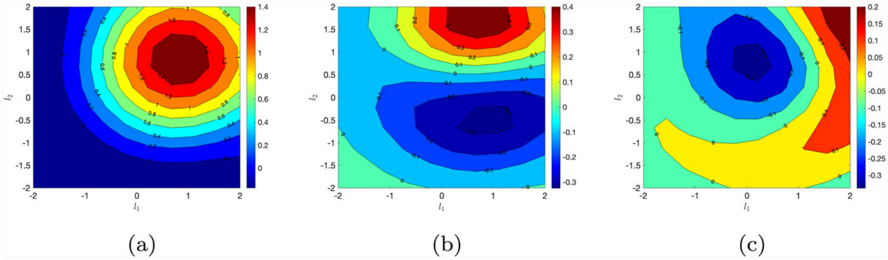

Fig. 5 The PMLW coefficients of $f(\mathbf{t}) = e^{\alpha_1 t_1 + \alpha_2 t_2}$, $\alpha_1, \alpha_2 \in \mathbb{R}$ with different l_1 and l_2. (a) Real-part. (b) $\mathbf{i}$-part. (c) $\mathbf{j}$-part

where $\phi_l = \tan^{-1}\left(\frac{\omega_{l,2}/b_2}{\omega_{l,1}/b_1}\right)$, $\boldsymbol{\zeta} = (s, \theta, \boldsymbol{\omega}_l)$, *and* $\boldsymbol{\omega}_l = (\omega_{l,1}, \omega_{l,2})$ *represents the linear canonical variables after the LCT has been applied in the* $\mathbf{l}$*-domain.*

Theorem 2 is directly derived from (29). Next, we present the implementation of the PMLW, as illustrated in Example 4.

Example 4 Consider the signal

$$f(\mathbf{t}) = e^{\alpha_1 t_1 + \alpha_2 t_2}, \quad \alpha_1, \alpha_2 \in \mathbb{R}. \tag{43}$$

Now, we examine the PMLW coefficients of $f(\mathbf{t})$ with $\alpha_1 = \alpha_2 = 0.5$, using the parametric monogenic extension of the DOG wavelet as introduced in Example 2.

- Let $s = 1.167$, $a_1 = a_2 = 1$, and $b_1 = b_2 = 2$. Compute the PMLW coefficients for different values of l_1 and l_2, as shown in Fig. 5. The main energy of the signal is concentrated around $l_1 = l_2 = 1$.
- Let $l_1 = l_2 = 1$, $a_1 = a_2 = 1$, and $b_1 = b_2 = 2$. Compute the PMLW coefficients for different values of s, as shown in Fig. 6. The signal's primary frequency characteristics remain stable over a large scale but exhibit local variations at smaller scales.
- Let $a_1 = a_2 = 1$, and $b_1 = b_2 = 2$. Compute the PMLW coefficients for different values of $l_1 = l_2$ and s, as shown in Fig. 7. Significant variations occur across different scales and translations, with the real part showing a strong central response.

The real-part mainly captures the global intensity or energy distribution of the signal, while the $\mathbf{i}$ and $\mathbf{j}$ components describe local variations of the signal along two distinct directions. This decomposition of the $\mathbf{i}$ and $\mathbf{j}$ parts is especially valuable for analyzing signal anisotropy, as it reveals the frequency characteristics across different directions.

If the linear canonical wavelet is radially symmetric, such that $\psi_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t}) = \psi^{(e)}_{\boldsymbol{\xi},\mathbf{A}}(\mathbf{t})$, the rotation has no significant effect on the definition of the CWT in (41). This symmetry implies that

$$sL_{\mathbf{A}}\{\psi_{\boldsymbol{\xi},\mathbf{A}}\}^{(e)*}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}) = sL_{\mathbf{A}}\{\psi_{\boldsymbol{\xi},\mathbf{A}}\}^{(e)}(s\boldsymbol{\omega}), \tag{44}$$

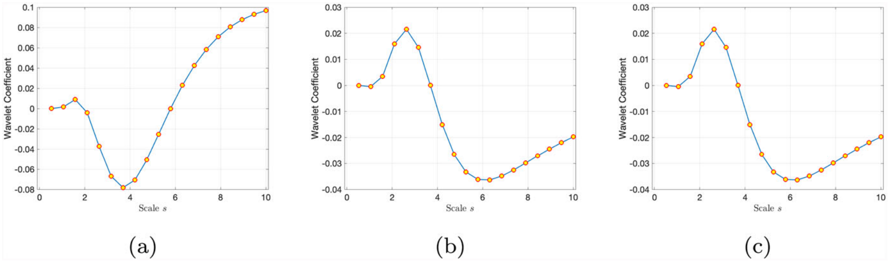

Fig. 6 The PMLW coefficients of $f(\mathbf{t}) = e^{\alpha_1 t_1 + \alpha_2 t_2}$, $\alpha_1, \alpha_2 \in \mathbb{R}$ with different s. (a) Real-part. (b) $\mathbf{i}$-part. (c) $\mathbf{j}$-part

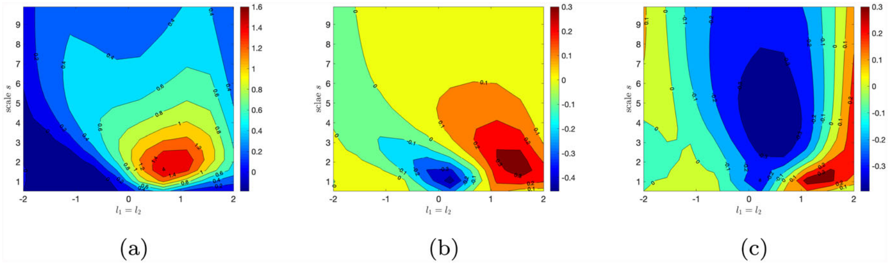

Fig. 7 The PMLW coefficients of $f(\mathbf{t}) = e^{\alpha_1 t_1 + \alpha_2 t_2}$, $\alpha_1, \alpha_2 \in \mathbb{R}$ with different $l_1 = l_2$ and s. (a) Real-part. (b) $\mathbf{i}$-part. (c) $\mathbf{j}$-part

where $(\cdot)^{(e)}$ denotes the even component. The LCT of the PMLW with this wavelet is given by

$$\begin{aligned} L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f)\right\}^{(e)}(\boldsymbol{\zeta}) =& s e^{\mathbf{i}\frac{d_1 s^2 \omega_{l,1}^2}{2b_1}} e^{\mathbf{i}\frac{d_2 s^2 \omega_{l,2}^2}{2b_2}} L_{\mathbf{A}}\{f\}^{(e)}(\boldsymbol{\omega}_l) \\ & \times L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^{(e)*}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}_l). \end{aligned} \tag{45}$$

This result follows from Lemma 2. The LCT of the PMLW with the parametric monogenic extension of this wavelet is given by

$$\begin{aligned} & L_{\mathbf{A}}\left\{w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi}, f)\right\}^{(e)}(\boldsymbol{\zeta}) = (1 + \cos(\phi_l - \theta) - \mathbf{k}\sin(\phi_l - \theta))\, L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}\right\}^{(e)}(\boldsymbol{\zeta}) \\ & = (1 + \cos(\phi_l - \theta) - \mathbf{k}\sin(\phi_l - \theta))\, s e^{\mathbf{i}\frac{d_1 s^2 \omega_{l,1}^2}{2b_1}} e^{\mathbf{i}\frac{d_2 s^2 \omega_{l,2}^2}{2b_2}} L_{\mathbf{A}}\{f\}^{(e)}(\boldsymbol{\omega}_l) \\ & \times L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2} e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^{(e)*}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}_l). \end{aligned} \tag{46}$$

Next, we define $\widetilde{f}_{-\theta}(\mathbf{t}) = f(\mathbf{R}_\theta \mathbf{t})$, and the parametric monogenic extension of $\widetilde{f}_{-\theta}(\mathbf{t})$ is given by

$$\widetilde{f}^{+}_{-\theta,\mathbf{A},\theta}(\mathbf{t}) = f(\mathbf{t}) - \mathbf{i}\widetilde{f}_{R_{\mathbf{A}}^{(1)},-\theta}(\mathbf{R}_{-\theta}\mathbf{t}) - \mathbf{j}\widetilde{f}_{R_{\mathbf{A}}^{(2)},-\theta}(\mathbf{R}_{-\theta}\mathbf{t})$$

$$=f(\mathbf{t})-\mathbf{i}\widetilde{f}_{R_{\mathbf{A}}^{(1)},-\theta,\theta}(\mathbf{t})-\mathbf{j}\widetilde{f}_{R_{\mathbf{A}}^{(2)},-\theta,\theta}(\mathbf{t}), \tag{47}$$

where $\widetilde{f}_{R_{\mathbf{A}}^{(k)},-\theta}(\mathbf{t})=R_{\mathbf{A}}^{(k)}\widetilde{f}_{-\theta}(\mathbf{t})$ for $k=1,2$. We now present the following theorem.

Theorem 3 *The LCT of the PMLW of the rotated parametric monogenic signal is given by*

$$L_{\mathbf{A}}\left\{w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi},\widetilde{f}^{+}_{-\theta,\mathbf{A},\theta})\right\}(\boldsymbol{\zeta})=2L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi},\widetilde{f}^{+}_{-\theta,\mathbf{A},\theta})\right\}(\boldsymbol{\zeta}). \tag{48}$$

Proof By definition, we can directly derive:

$$\begin{aligned}
&L_{\mathbf{A}}\left\{w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi},\widetilde{f}^{+}_{-\theta',\mathbf{A},\theta'})\right\}(\boldsymbol{\zeta})\\
&=L_{\mathbf{A}}\left\{w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi},f)\right\}(\boldsymbol{\zeta})-\mathbf{i}L_{\mathbf{A}}\left\{w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi},\widetilde{f}_{R_{\mathbf{A}}^{(1)},-\theta',\theta'})\right\}(\boldsymbol{\zeta})\\
&\quad-\mathbf{j}L_{\mathbf{A}}\left\{w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi},\widetilde{f}_{R_{\mathbf{A}}^{(2)},-\theta',\theta'})\right\}(\boldsymbol{\zeta})\\
&=(1+\cos(\phi_l-\theta)-\mathbf{k}\sin(\phi_l-\theta))\,L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi},f)\right\}(\boldsymbol{\zeta})\\
&\quad-\mathbf{i}\,(1+\cos(\phi_l-\theta)-\mathbf{k}\sin(\phi_l-\theta))\left(\mathbf{i}\cos(\phi_l-\theta')\right)L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi},f)\right\}(\boldsymbol{\zeta})\\
&\quad-\mathbf{j}\,(1+\cos(\phi_l-\theta)-\mathbf{k}\sin(\phi_l-\theta))\left(\mathbf{i}\sin(\phi_l-\theta')\right)L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi},f)\right\}(\boldsymbol{\zeta})\\
&=(1+\cos(\phi_l-\theta)-\mathbf{k}\sin(\phi_l-\theta)+\cos(\phi_l-\theta')+\cos(\phi_l-\theta)\cos(\phi_l-\theta')\\
&-\mathbf{k}\sin(\phi_l-\theta)\cos(\phi_l-\theta')-\mathbf{k}\sin(\phi_l-\theta')-\mathbf{k}\cos(\phi_l-\theta)\sin(\phi_l-\theta')\\
&\quad+\sin(\phi_l-\theta)\sin(\phi_l-\theta'))L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi},f)\right\}(\boldsymbol{\zeta}).
\end{aligned} \tag{49}$$

Therefore, it follows that

$$\begin{aligned}
&L_{\mathbf{A}}\left\{w^{+}_{\psi,\mathbf{A}}(\boldsymbol{\xi},\widetilde{f}^{+}_{-\theta,\mathbf{A},\theta})\right\}(\boldsymbol{\zeta})\\
&=(2+2\cos(\phi_l-\theta)-2\mathbf{k}\sin(\phi_l-\theta))L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi},f)\right\}(\boldsymbol{\zeta})\\
&=(2+2\cos(\phi_l-\theta)-2\mathbf{k}\sin(\phi_l-\theta))se^{\mathbf{i}\frac{d_1s^2\omega_{l,1}^2}{2b_1}}e^{\mathbf{i}\frac{d_2s^2\omega_{l,2}^2}{2b_2}}L_{\mathbf{A}}\left\{f\right\}(\boldsymbol{\omega}_l)\\
&\quad\times L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2}e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^{*}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}_l)\\
&=(2+2\cos(\phi_l-\theta)-2\mathbf{k}\sin(\phi_l-\theta))se^{\mathbf{i}\frac{d_1s^2\omega_{l,1}^2}{2b_1}}e^{\mathbf{i}\frac{d_2s^2\omega_{l,2}^2}{2b_2}}L_{\mathbf{A}}\left\{\widetilde{f}_{-\theta,\mathbf{A},\theta}\right\}(\boldsymbol{\omega}_l)\\
&\quad\times L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2}e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^{*}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}_l)\\
&=2se^{\mathbf{i}\frac{d_1s^2\omega_{l,1}^2}{2b_1}}e^{\mathbf{i}\frac{d_2s^2\omega_{l,2}^2}{2b_2}}L_{\mathbf{A}}\left\{\widetilde{f}^{+}_{-\theta,\mathbf{A},\theta}\right\}(\boldsymbol{\omega}_l)L_{\mathbf{A}}\left\{\psi e^{-\mathbf{i}\frac{a_1}{2b_1}(\cdot)^2}e^{-\mathbf{i}\frac{a_2}{2b_2}(\cdot)^2}\right\}^{*}(s\mathbf{R}_{-\theta}\boldsymbol{\omega}_l)\\
&=2L_{\mathbf{A}}\left\{w_{\psi,\mathbf{A}}(\boldsymbol{\xi},\widetilde{f}^{+}_{-\theta,\mathbf{A},\theta})\right\}(\boldsymbol{\zeta}).
\end{aligned} \tag{50}$$

$\square$

6 Image Denoising Based on PMLW

In this section, we discuss the application of PMLW to the image denoising task. For an input image $f(m,n)$, the first step is to apply the linear canonical modulation to embed the image into the transformed domain using the following phase transformation [14]

$$f_{\mathbf{A}}(m,n) = f(m,n)e^{\mathbf{i}\left(\frac{a_1}{2b_1}m^2+\frac{a_2}{2b_2}n^2\right)}. \tag{51}$$

After modulation, we apply the rotational monogenic discrete wavelet transform (DWT) to obtain the four subbands: the parametric monogenic linear canonical domain low-frequency subband $LL_{\mathbf{A}}^{+} = LL_{\mathbf{A}} + \mathbf{i}LL_{\mathbf{A}}^{(1)} + \mathbf{j}LL_{\mathbf{A}}^{(2)}$, the horizontal high-frequency subband $HL_{\mathbf{A}}^{+} = HL_{\mathbf{A}} + \mathbf{i}HL_{\mathbf{A}}^{(1)} + \mathbf{j}HL_{\mathbf{A}}^{(2)}$, the vertical high-frequency subband $LH_{\mathbf{A}}^{+} = LH_{\mathbf{A}} + \mathbf{i}LH_{\mathbf{A}}^{(1)} + \mathbf{j}LH_{\mathbf{A}}^{(2)}$, and the diagonal high-frequency subband $HH_{\mathbf{A}}^{+} = HH_{\mathbf{A}} + \mathbf{i}HH_{\mathbf{A}}^{(1)} + \mathbf{j}HH_{\mathbf{A}}^{(2)}$. The reconstruction process follows an inverse transformation by first applying the inverse rotational monogenic DWT to recover the modulated signal $f_{\mathbf{A}}^{\mathrm{rec}}(m,n)$. Finally, the inverse modulation restores the image into the spatial domain

$$f^{\mathrm{rec}}(m,n) = f_{\mathbf{A}}^{\mathrm{rec}}(m,n)e^{-\mathbf{i}\left(\frac{a_1}{2b_1}m^2+\frac{a_2}{2b_2}n^2\right)}. \tag{52}$$

The denoising process in PMLW is performed by applying thresholding techniques to the high-frequency subbands while preserving the low-frequency subband to maintain the structural details of the image. Since noise is more prominent in the high-frequency components, a soft-thresholding function is applied to suppress noise while retaining useful image details

$$\widetilde{HL_{\mathbf{A}}} = |LL_{\mathbf{A}}^{+}|, \tag{53}$$

$$\widetilde{HL_{\mathbf{A}}} = \eta(|HL_{\mathbf{A}}, \lambda)| + \beta\left(|\eta(HL_{\mathbf{A}}^{(1)}, \lambda)| + |\eta(HL_{\mathbf{A}}^{(2)}, \lambda)|\right), \tag{54}$$

$$\widetilde{LH_{\mathbf{A}}} = \eta(|LH_{\mathbf{A}}, \lambda)| + \beta\left(|\eta(LH_{\mathbf{A}}^{(1)}, \lambda)| + |\eta(LH_{\mathbf{A}}^{(2)}, \lambda)|\right), \tag{55}$$

$$\widetilde{HH_{\mathbf{A}}} = \eta(|HH_{\mathbf{A}}, \lambda)| + \beta\left(|\eta(HH_{\mathbf{A}}^{(1)}, \lambda)| + |\eta(HH_{\mathbf{A}}^{(2)}, \lambda)|\right), \tag{56}$$

where $\eta(x,\lambda)$ is the soft-thresholding function

$$\eta(x,\lambda) = \mathrm{sign}(x)\max(|x|-\lambda, 0), \tag{57}$$

with λ as the threshold parameter, and β is a weighting parameter that controls the contribution of the directional enhancement. After denoising, the modified subbands $\widetilde{LL_{\mathbf{A}}}$, $\widetilde{HL_{\mathbf{A}}}$, $\widetilde{LH_{\mathbf{A}}}$, $\widetilde{HH_{\mathbf{A}}}$ are used in the inverse PMLW process to reconstruct the denoised image.

To evaluate the effectiveness of the proposed PMLW-based denoising approach, we conduct comparative experiments with two conventional wavelet-based denoising methods: DWT and LCWT. The key difference between these methods is that

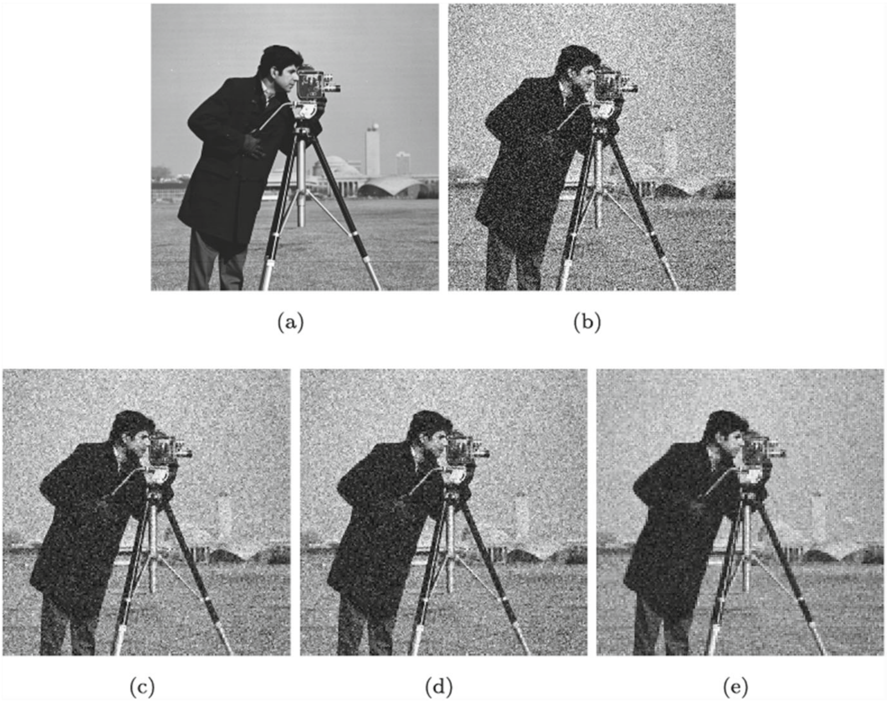

Fig. 8 Image denoising results obtained using different methods: (a) Original image. (b) Noisy image (with a noise standard deviation of 0.02). (c) DWT (PSNR = 20.46). (d) LCWT (PSNR = 21.04). (e) PMLW (PSNR = 24.21)

both DWT and LCWT only apply soft-thresholding to the wavelet coefficients, without utilizing the additional parametric monogenic components **i** and **j** for directional enhancement. In contrast, PMLW incorporates these components to enhance the structural details in the image while reducing noise. In the experiments, we set the denoising parameters as follows: $\lambda = 0.1$, $\beta = 0.05$, $\theta = \pi/8$, and transformation parameters $a_1 = a_2 = 1, b_1 = b_2 = 2$. The noise levels are varied to investigate the robustness of different methods under different Gaussian noise intensities. The peak signal-to-noise ratio (PSNR) is used as quantitative evaluation metrics [34]. When the noise standard deviation is 0.02, the results are shown in Fig. 8. Fig. 9 (a) shows that PMLW consistently achieves higher PSNR than DWT and LCWT, especially at higher noise levels, demonstrating its robustness in severe noise conditions. Fig. 9 (b) illustrates how the rotation angle θ affects denoising performance. While PSNR remains relatively stable for DWT and LCWT across different angles, PMLW exhibits a gradual decline in PSNR as θ increases. This suggests that PMLW benefits from directionality at lower angles but may experience performance degradation with larger rotations due to directional information dispersion. The results demonstrate that incorporating monogenic analysis into the linear canonical wavelet framework improves both noise suppression and structural preservation. The additional directional information allows for a more

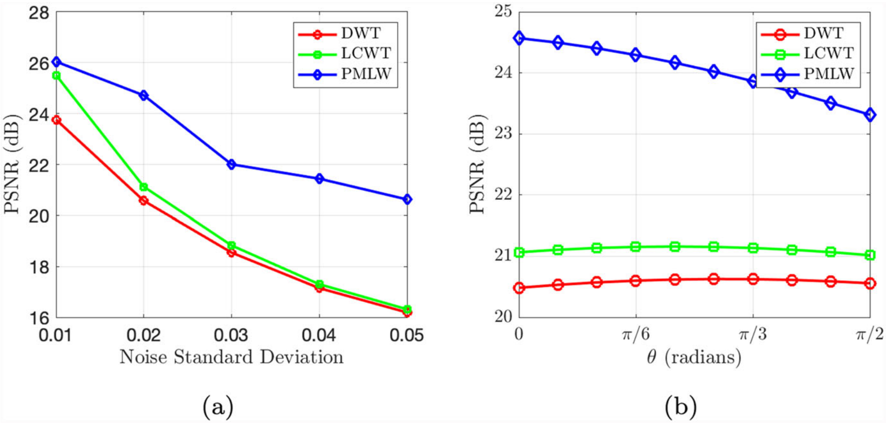

Fig. 9 Denoising performance of different methods under varying parameters: (a) Different noise standard deviations. (b) Different rotation angles (θ)

adaptive denoising process, making PMLW a superior choice for image restoration tasks in noisy environments.

7 Conclusion

This paper develops the two-dimensional linear canonical wavelet transform (LCWT) and extends it by introducing the parametric monogenic linear canonical wavelet transform (PMLW). The LCWT is formulated with rotation factors to address multi-directional analysis, and a novel parametric wavelet model is constructed by integrating the parametric Riesz transform and monogenic signal. The properties and definitions of the LCWT and PMLW are systematically analyzed, providing a detailed framework for the implementation and application of these transforms. Furthermore, the application of PMLW in image denoising is explored, demonstrating its potential to preserve structural details while effectively suppressing noise. In future work, PMLW will be further investigated in terms of parameter selection, the uncertainty principle, and its broader theoretical implications.

Acknowledgements This work was supported by the National Natural Science Foundation of China [No. 62171041].

Author Contributions Jian-Yi Chen: Conceptualization, Original draft. Bing-Zhao Li: Conceptualization, Funding acquisition, Review and editing.

Data Availability Not applicable.

Declarations

Conflict of interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Materials availability Not applicable.

Code availability Not applicable.

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